

Vipul Shrivastav

Software Developer | Python · Backend · Computer Vision · SQL
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PROFILE SUMMARY

2026 CSE graduate currently interning as a Software Developer at SmarDen Automation where I have built real Python applications for clients like Blinkit and Maruti Suzuki. I have worked on backend development (Django, Node.js, REST APIs) and computer vision projects. Good understanding of OOP, DBMS, SQL, and OS concepts. Looking to join a team where I can contribute and keep learning.

EXPERIENCE

Software Developer Intern — SmarDen Automation, Haryana

June 2025 – Present

- ▶ Built a real-time product counting system for Blinkit's industrial conveyor belt using Python and OpenCV — production-ready POC designed for Raspberry Pi deployment.
- ▶ Engineered automatic lighting-change detection that pauses/resumes counting during illumination events, eliminating false counts in live production.
- ▶ Trained a custom YOLOv11 model on a self-annotated dataset for Maruti Suzuki achieving 91.9% mAP · 99% Precision · 88.5% Recall · 93.5% F1.
- ▶ Managed full ML pipeline: data collection → Roboflow annotation → 4x augmentation → Colab T4 GPU training → inference; reduced annotation effort by ~80% via active learning.
- ▶ Built clean, modular codebases (detector.py, counter.py, CartonBoxDetector class) following production-grade software design principles.

PROJECTS

Conveyor Belt Product Counting System — Blinkit | Python · OpenCV · MOG2 · NumPy

- ▶ Implemented MOG2 background subtraction with morphological operations to detect and track moving products across a virtual counting line.
- ▶ ROI-based processing for real-time performance; supports USB camera, IP webcam, and video file — zero code changes across input types.

Carton Box Detection System — Maruti Suzuki | Python · YOLOv11 · OpenCV · Roboflow

- ▶ Custom-trained YOLOv11 model with modular architecture (CartonBoxDetector class) supporting image, video, and live camera streams.
- ▶ Designed for future extension — conveyor belt tracking, object counting, line crossing — without modifying input stream scripts.

TECHNICAL SKILLS

Languages: Python · C · JavaScript

Backend: Django · Node.js · Express.js · REST APIs · FastAPI (learning)

Frontend: HTML · CSS · JavaScript · React (basics)

Databases: MySQL · MongoDB

Computer Vision / ML: OpenCV · YOLOv11 · Roboflow · NumPy

Tools & Concepts: Git · GitHub · Postman · VS Code · OOPs · DBMS · OS · CN · SDLC · DSA

EDUCATION

B.Tech — Computer Science & Engineering

Maharana Pratap Group of Institutions, Kanpur

2022 – 2026

CGPA: 7.4 / 10

CERTIFICATIONS

Problem Solving — HackerRank | SQL 3-Star Badge — HackerRank | MongoDB Basics — GeeksforGeeks | LeetCode (Arrays · Strings · Hashmaps · Two Pointers · Sliding Window)